

(2023) 期中考试题参考答案.

P1

Proof of 1. (a) 正变换 $\begin{cases} u = \frac{x}{y} \\ v = x^2 + y^2 \end{cases} : (x, y) \rightarrow H(u, v) = (u, v)$

则 $H(u, v)$ 有两支 -- 反变换:

$$H_1^{-1}(u, v) = (x_1, y_1) : \begin{cases} x_1(u, v) = u \sqrt{\frac{v}{1+u^2}} \\ y_1(u, v) = \sqrt{\frac{v}{1+u^2}} \end{cases}$$

$$H_2^{-1}(x, y) = (x_2, y_2) : \begin{cases} x_2(u, v) = -u \sqrt{\frac{v}{1+u^2}} \\ y_2(u, v) = -\sqrt{\frac{v}{1+u^2}} \end{cases}$$

$$|J_1(u, v)| = \left| \frac{\partial(x_1, y_1)}{\partial(u, v)} \right| = \frac{1}{2(1+u^2)} = |J_2(u, v)| = \left| \frac{\partial(x_2, y_2)}{\partial(u, v)} \right|.$$

$$\begin{aligned} P_U, V(u, v) &= P_{3_1}(x_1(u, v)) P_{3_2}(y_1(u, v)) |J_1(u, v)| + \\ &\quad P_{3_1}(x_2(u, v)) P_{3_2}(y_2(u, v)) |J_2(u, v)| \\ &= \frac{1}{2\pi} e^{-\frac{1}{2}v} \left(\frac{1}{2(1+u^2)} + \frac{1}{2(1+u^2)} \right) \\ &= \frac{1}{\pi} \frac{1}{(1+u^2)} \frac{1}{2} e^{-\frac{1}{2}v}, \end{aligned}$$

$$\text{where } P_{3_1}(x) = P_{3_2}(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2},$$

$$\Rightarrow P_U(u) = \frac{1}{\pi} \frac{1}{1+u^2}, \quad P_V(v) = \frac{1}{2} e^{-\frac{1}{2}v}.$$

$$\Rightarrow U \text{ and } V \text{ are indep.}, \quad U \sim C(1, 0), \quad V \sim \chi^2_{(2)}$$

$$(b) \text{ As } \left\{ \frac{z_{2k+1}}{z_{2k}}, k \geq 1 \right\} \text{ are indep, and (a)} \Rightarrow \frac{z_{2k+1}}{z_{2k}} \sim C(1, 0),$$

$$\text{利用加法定理, } Z = \sum_{k=1}^n \frac{z_{2k+1}}{z_{2k}} \sim C(n, 0)$$

$$\Rightarrow Z \text{ has density funct. } P_Z(z) = \frac{1}{\pi} \frac{n}{n^2 + z^2}.$$

(C) set $X = \sum_{k=1}^m z_k^2, Y = \sum_{k=1}^n \eta_k^2 \Rightarrow X \sim \chi^2(m), Y \sim \chi^2(n)$ $\boxed{P_2}$

$Z \triangleq \frac{X}{Y/n} = \frac{n}{m} W \sim F(m, n)$. 并设 Z 的密度函数

为 $P_Z(z) \triangleq \frac{\Gamma(\frac{m+n}{2})}{\Gamma(\frac{m}{2})\Gamma(\frac{n}{2})} m^{\frac{m}{2}} n^{\frac{n}{2}} z^{\frac{m}{2}-1} (n+mx)^{-\frac{1}{2}(m+n)}$ ($F_{m,n}$ 的分布密度函数)

则 $W = \frac{m}{n} Z$ 为 Z 的线性变换: $y \triangleq \frac{m}{n} x \Rightarrow x = \frac{n}{m} y, \frac{dx}{dy} = \frac{n}{m}$

$\Rightarrow W$ has density function:

$$P_W(y) = P_Z(\frac{n}{m}y) \frac{n}{m} = \frac{\Gamma(\frac{m+n}{2})}{\Gamma(\frac{n}{2})\Gamma(\frac{m}{2})} y^{\frac{m}{2}-1} (1+y)^{-\frac{n+m}{2}}, y > 0.$$

(d) $E\{W\} = E\{\frac{m}{n}Z\}$

$$= \frac{m}{n} E\{Z\} = \frac{m}{n} \frac{n}{n-2} = \frac{m}{n-2}.$$

2 题

Proof of 2. (a) do $P_{Z_1}(x) = e^{-x} I_{\{x>0\}}, y=2x, x=\frac{1}{2}y, \frac{dx}{dy} = \frac{1}{2}$.

$$\Rightarrow P_Y(y) = P_{Z_1}(\frac{1}{2}y) \left| \frac{dx}{dy} \right| = \frac{1}{2} e^{-\frac{1}{2}y} I_{[0, \infty)}(y)$$

$$\Rightarrow 2Z_1 \sim \Gamma(\frac{1}{2}, 1)$$

(ii) $U \triangleq Z_1 + Z_2, V = \frac{Z_1}{Z_1 + Z_2}, H(u, v) \triangleq (u, v):$ $u = x+y, v = \frac{x}{x+y}$

$\Rightarrow H$ 有逆变换: $x(u, v) = uv, y(u, v) = u(1-v) \Rightarrow |J(u, v)| = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| = u,$

(利用雅可比)

$$\Rightarrow P_{(U, V)}(u, v) = P_{Z_1}(x(u, v)) P_{Z_2}(y(u, v)) |J(u, v)|$$

$$= u e^{-u} I_{\{u>0\}} \cdot I_{[0, 1]}(v) = P_U(u) P_V(v)$$

$$\Rightarrow P_U(u) = u e^{-u} I_{[0, \infty)}(u), P_V(v) = I_{[0, 1]}(v) = \begin{cases} 1, & v \in [0, 1] \\ 0, & v \notin [0, 1] \end{cases}$$

$$\Rightarrow V \sim U[0, 1].$$

$$(b) \quad (i) \quad X = e^{-\eta_3 (\log \frac{1}{\eta_1} + \log \frac{1}{\eta_2})}$$

P3

$$\triangleq e^{-3\eta}, \quad 3 \triangleq \eta_3, \quad \eta \triangleq \log \frac{1}{\eta_1} + \log \frac{1}{\eta_2}$$

$\Rightarrow 3 \sim U_{[0,1]}, \quad \eta \sim T(1,2)$ (as $\log \frac{1}{\eta_i} \sim T(1,1), i=1,2$, they are indep.).

$$P_3(x) = \begin{cases} 1, & x \in [0,1], \\ 0, & x \in [0,1]^c, \end{cases} \quad P_\eta(y) = y e^{-y}$$

$$P_{(3,\eta)}(x,y) = y e^{-y} I_{[0,1]}^{(x)} I_{[0,\infty)}^{(y)}$$

作变换 $\begin{cases} u \triangleq e^{-3\eta} \\ v \triangleq \eta \end{cases} \quad \begin{cases} u = e^{-xy} \\ v = y \end{cases} \Rightarrow |J(u,v)| = \frac{1}{uv}$

$\Rightarrow (U,V)$ has joint density function:

$$P_{(U,V)}(u,v) = 1 \times v e^{-v} \frac{1}{uv} I_{[e^{-v},1]}^{(u)} I_{[0,\infty)}^{(v)}$$

$$\Rightarrow P_U(u) = \int_{\mathbb{R}} P_{(U,V)}(u,v) dv = \int_{\log \frac{1}{u}}^{\infty} \frac{1}{u} e^{-v} dv = 1, \quad u \in [0,1].$$

$$\Rightarrow U = X \sim U_{[0,1]}$$

$$(ii) \quad \text{As } Y = \frac{Z_X}{Z_m} \Big/ \frac{Z_Y}{Z_{(n-m)}}, \quad Z_X \triangleq \sum_{k=1}^m Z_{3,k} \sim T(\frac{1}{2}, \frac{2m}{2})$$

$$Z_Y \triangleq \sum_{k=m+1}^n Z_{3,k} \sim T(\frac{1}{2}, \frac{2(n-m)}{2}) \quad (\text{利用 (a) 的独立性}).$$

and X 与 Y 独立, 则 $F(m,n)$ 的分布,

$$Y \sim F(Z_m, Z_{(n-m)}).$$

$$(C) \quad EY = \frac{Z_{(n-m)}}{Z_{(n-m)} - 2} = \frac{n-m}{n-m-1}.$$

3 题.

Proof of (1): (b) Let $B \triangleq (0, 1)$, $B_1 \triangleq (-1, 1)$, then P_4
 $B, B_1 \in \mathcal{B}(R)$. As $P\{\omega: |\xi(\omega)| \in B\} \in (0, 1)$ and

$\{\omega: |\xi(\omega)| \in B\} \supset \{\omega: \xi(\omega) \in B_1\}$, we have

$$P\{\xi(\omega) \in B_1, |\xi(\omega)| \in B\} = P\{\omega: \xi(\omega) \in B_1\}$$

$$\neq P\{\omega: \xi(\omega) \in B_1\} P\{\omega: |\xi(\omega)| \in B\}.$$

i.e., ξ and $|\xi|$ are ^{not} indep.

Proof of (2): As $E(\xi) = 0$, $E(\xi|\xi_1) = 0$, we have

$$E(\xi) \cdot E(|\xi|) = E(\xi|\xi_1)$$

$\Rightarrow \xi$ and $|\xi|$ are unrelated r.v.'s. #

Proof of (2)

(a) Taking $B = (-1, 1)$, $B_1 = (0, 1)$. Then $B, B_1 \in \mathcal{B}(R)$ and

$$P\{\omega: \xi(\omega) \in B\} \in (0, 1)$$

As $\{\xi \in B\} \supset \{\xi^2 \in B_1\}$, we have

$$P\{\xi \in B, \xi^2 \in B_1\} = P\{\xi \in B_1\} \neq P\{\xi \in B\} P\{\xi^2 \in B_1\}$$

$\Rightarrow \xi$ and ξ^2 are not indep.

(b) as $E\xi = E\xi^3 = 0$, $E\xi \cdot E\xi^3 = E\xi^3 = 0$,

ξ and ξ^3 are unrelated r.v.'s.

4. 題

Proof. (a) As $\{A_k, k=1, \dots\}$ is G-meas, γ is G-meas.

(b) $\mathcal{C} \triangleq \{A_k, \phi: k \geq 1\}$. $\mathcal{C}$ is π -system $\Rightarrow$

① $\sigma(\mathcal{C}) = \lambda(\mathcal{C}) = \mathcal{G}$. 2

$$\tilde{\mathcal{G}} = \{A \in \mathcal{G} \mid \int_A \xi \, d\mu = \int_A \gamma \, d\mu\}.$$

P5

Using $E\{\xi \mid \mathcal{A}_n\} = \frac{E\{\xi I_{\mathcal{A}_n}\}}{P(\mathcal{A}_n)}$, $\int_{\mathcal{A}_n} \gamma \, d\mu = \int_{\mathcal{A}_n} E\{\xi \mid \mathcal{A}_n\} \, d\mu$ (2) = 4

$$= E\{\xi I_{\mathcal{A}_n}\} = \int_{\mathcal{A}_n} \xi \, d\mu, \text{ thus, } \mathcal{A}_n \in \tilde{\mathcal{G}} \Rightarrow$$

$$\tilde{\mathcal{G}} \supseteq \mathcal{G}. \text{ and } \tilde{\mathcal{G}} \text{ is a-system. } \textcircled{3} = \textcircled{*} \textcircled{1} + \textcircled{3}_2 + \textcircled{3}_3 = 4$$

Indeed, (i) $\Omega \in \tilde{\mathcal{G}}$ is clear. = 1 + 1 + 2

$$\text{(ii) } B \supset A, B, A \in \tilde{\mathcal{G}}: \left. \begin{aligned} \int_B \xi \, d\mu &= \int_B \gamma \, d\mu \\ \int_A \xi \, d\mu &= \int_A \gamma \, d\mu \end{aligned} \right\} \Rightarrow$$

$$\begin{aligned} \int_{B-A} \xi \, d\mu &= \int_B \xi \, d\mu - \int_A \xi \, d\mu = \int_B \gamma \, d\mu - \int_A \gamma \, d\mu \\ &= \int_{B-A} \gamma \, d\mu. \end{aligned}$$

$$\Rightarrow B-A \in \tilde{\mathcal{G}}$$

2 (iii) $\{A_n \uparrow A, A_n \in \tilde{\mathcal{G}}\}$. As $\xi I_{A_n} \rightarrow \xi I_A, \gamma I_{A_n} \rightarrow \gamma I_A$

and $|\xi I_{A_n}| \leq |\xi|, |\gamma I_{A_n}| \leq |\gamma|$, $\xi, \gamma \in L^1(\mu, \mathbb{R})$

Using DCT, and $\int_{A_n} \xi \, d\mu = \int_{A_n} \gamma \, d\mu$, letting $n \rightarrow \infty$

we have $\int_A \xi \, d\mu = \int_A \gamma \, d\mu \Rightarrow A \in \tilde{\mathcal{G}}$.

Based on (i), (ii) and (iii), $\tilde{\mathcal{G}}$ is a-system $\Rightarrow$

$$\tilde{\mathcal{G}} \supseteq \mathcal{A}(\mathcal{C}) = \mathcal{O}(\mathcal{C}) = \mathcal{G} \Rightarrow \mathcal{G} = \tilde{\mathcal{G}} \Rightarrow (b) \text{ holds.}$$

(c) Let γ_1 and γ_2 satisfy (a) and (b) $\Rightarrow \underline{\gamma_1 \text{ and } \gamma_2 \text{ are}}$

(5) $\mathcal{G}$ -meas., and $\int_A \gamma_1 \, d\mu = \int_A \gamma_2 \, d\mu \Rightarrow \gamma_1 \stackrel{\text{a.s.}}{=} \gamma_2$.
 $\forall A \in \mathcal{G}$

#

Proof: 設 F and F_1 為 two p.d.f., F and F_1 are the l.s meas. generated by F and F_1 , respectively.

set $(R, \mathcal{B}(R)) \triangleq (\Omega_i, \mathcal{F}_i)$, $i=1, 2$.

$$X_i(\omega_i) = \omega_i, \quad \omega_i \in \Omega_i.$$

$$P_i(\cdot) \triangleq F_i(\cdot), \quad P_i(\cdot) = F_i(\cdot) \Rightarrow P_i(\omega_i: X_i(\omega_i) \leq x) = F_i(x)$$

then $(\Omega_i, \mathcal{F}_i, P_i)$ are two P.S. (10) $\forall x \in R$,

$$\text{set } X(\omega) = X_1(\omega_1), \quad \omega = (\omega_1, \omega_2) \in \Omega \triangleq \Omega_1 \times \Omega_2$$

$$Y(\omega) = X_2(\omega_2), \quad \omega = (\omega_1, \omega_2).$$

$$\mathcal{F} \triangleq \mathcal{F}_1 \cup \mathcal{F}_2, \quad P(\cdot) = (P_1 \times P_2)(\cdot). \quad \text{5}$$

Then $\forall B_1, B_2 \in \mathcal{B}(R) = \mathcal{F}_i$, $i=1, 2$,

$$P(\omega: X_1(\omega) \in B_1, Y(\omega) \in B_2)$$

$$= (P_1 \otimes P_2)((\omega_1, \omega_2) \in B_1 \times B_2)$$

$$= P_1(\omega_1: \omega_1 \in B_1) P_2(\omega_2: \omega_2 \in B_2)$$

$$= P_1(\omega_1: X_1(\omega_1) \in B_1) P_2(\omega_2: X_2(\omega_2) \in B_2) \quad \text{4}$$

$$\text{取 } B_2 = \Omega_2 \Rightarrow P(\omega: X_1(\omega) \in B_1) = P_1(\omega_1: X_1(\omega_1) \in B_1)$$

$$\text{Similarly, } P(\omega: Y(\omega) \in B_2) = P_2(\omega_2: X_2(\omega_2) \in B_2)$$

$$\Rightarrow P(\omega: X_1(\omega) \in B_1, Y(\omega) \in B_2) = P(\omega: X_1(\omega) \in B_1) P(\omega: Y(\omega) \in B_2)$$

i.e., X and Y are indep. under P . 5 = 2 + 3

$$\text{取 } B_2 = \Omega \text{ and } B_1 = (-\infty, x),$$

$$P\{\omega: X_1(\omega) \leq x\} = P_1(\omega_1: X_1(\omega_1) \leq x) = F(x), \quad \forall x \in R$$

6 題

Proof of (1): $\Rightarrow$ As z and y are indep,

$$(5) \quad E\{|z+y|^p\} = \int_{\mathbb{R}} E\{|z+y|^p\} dF_y(y) < +\infty$$

$$\Rightarrow \exists y_0 \in \mathbb{R} \text{ st } E\{|z+y_0|^p\} < +\infty.$$

$$\text{Using Cr-ineq, } E\{|z|^p\} \leq \max\{2^p, 1\} \{E\{|y_0|^p\}, E\{|z+y_0|^p\}\} < +\infty.$$

$$\text{Similarly, } E\{|y|^p\} < +\infty.$$

$$(5) \quad \Leftarrow \text{Based on Cr-ineq, } E\{|z+y|^p\} \leq \max\{1, 2^p\} \{E\{|z|^p\}, E\{|y|^p\}\} < +\infty.$$

Proof of (2). As z and y are indep,

$$(5) \quad E\{|z+y|^p\} = \int_{\mathbb{R}} E\{|x+y|^p\} dF_z(x). \quad (3*)$$

Using $g(x) = |x|^p$ ($p \geq 1$) is convex function, and Jensen's ineq, we have

$$(5) \quad E\{|x+y|^p\} \geq |E\{x\} + E\{y\}|^p = |x|^p,$$

thus, using (3*) yields

$$E\{|z+y|^p\} \geq \int_{\mathbb{R}} |x|^p dF_z(x) = E\{|z|^p\}. \quad \#$$